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(54) **UNMANNED UNDERWATER VEHICLE WITH VARIABLE-GEOMETRY HULL**

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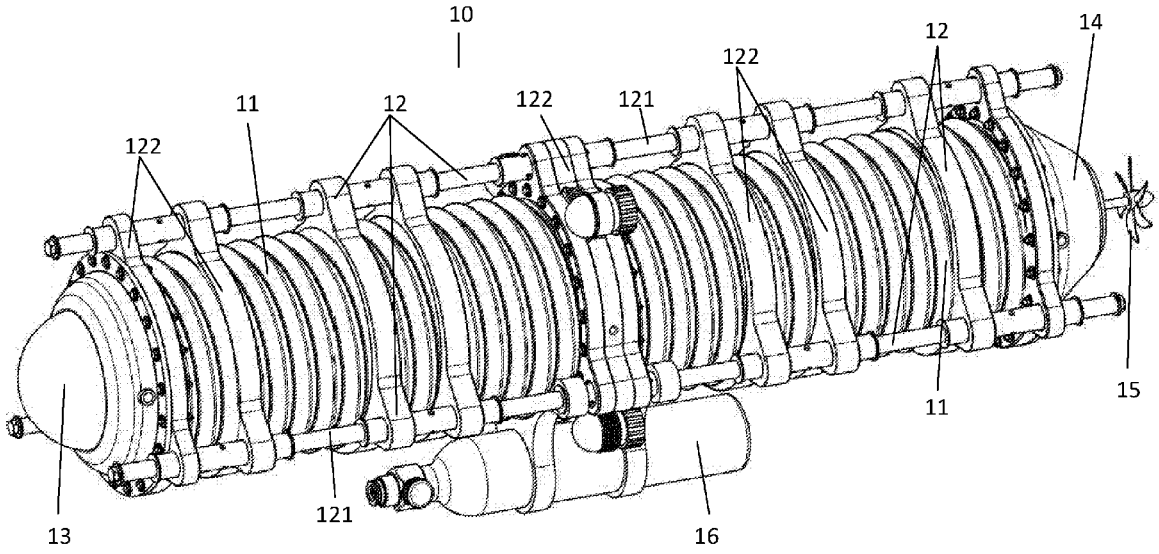
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(57) **ABSTRACT**

Unmanned underwater vehicle with variable-geometry internally pressurized hull that enables the underwater vehicle to submerge/emerge and change submersion depth by varying hull's buoyancy and not the vehicle weight.



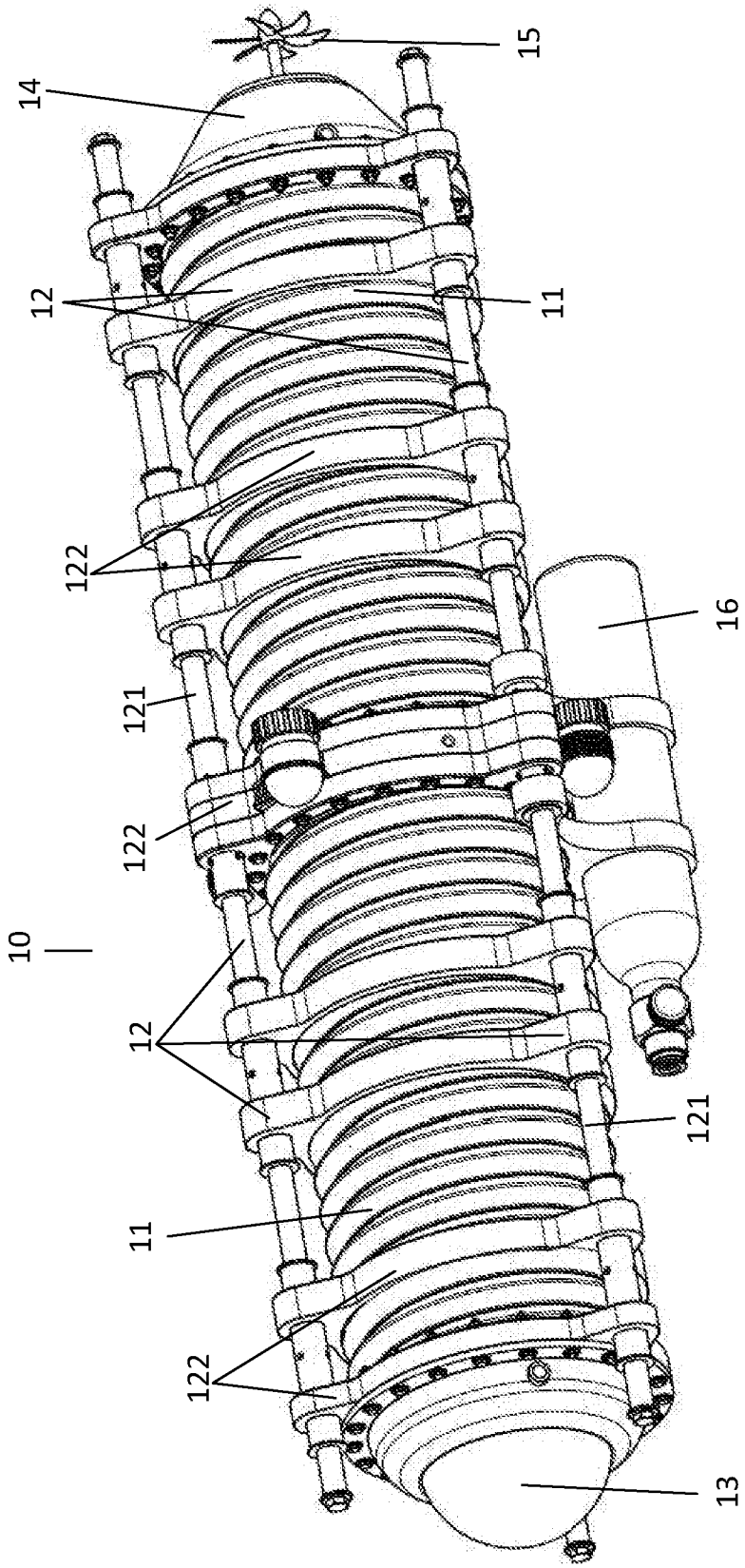


Fig. 1

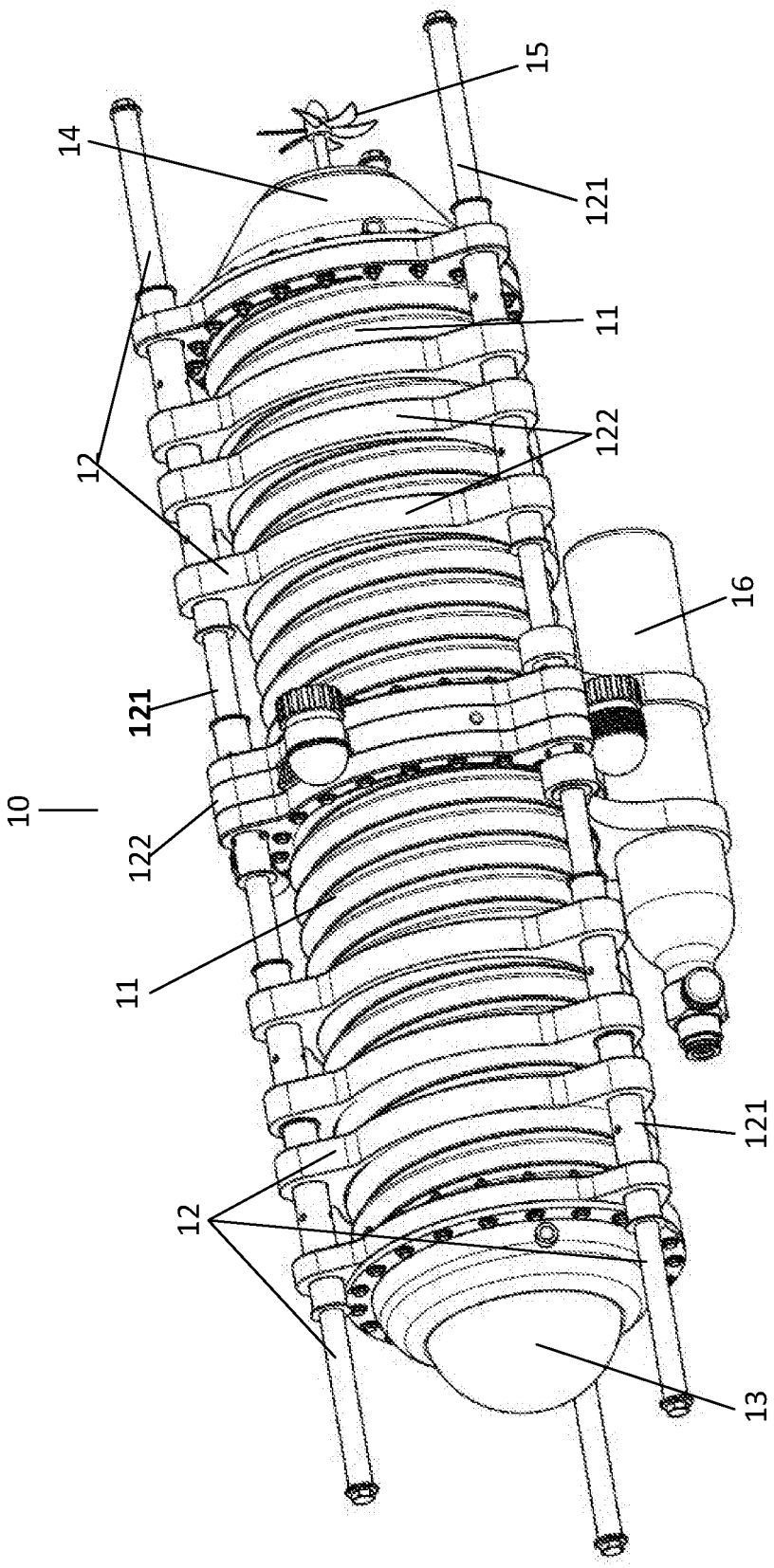


Fig. 2

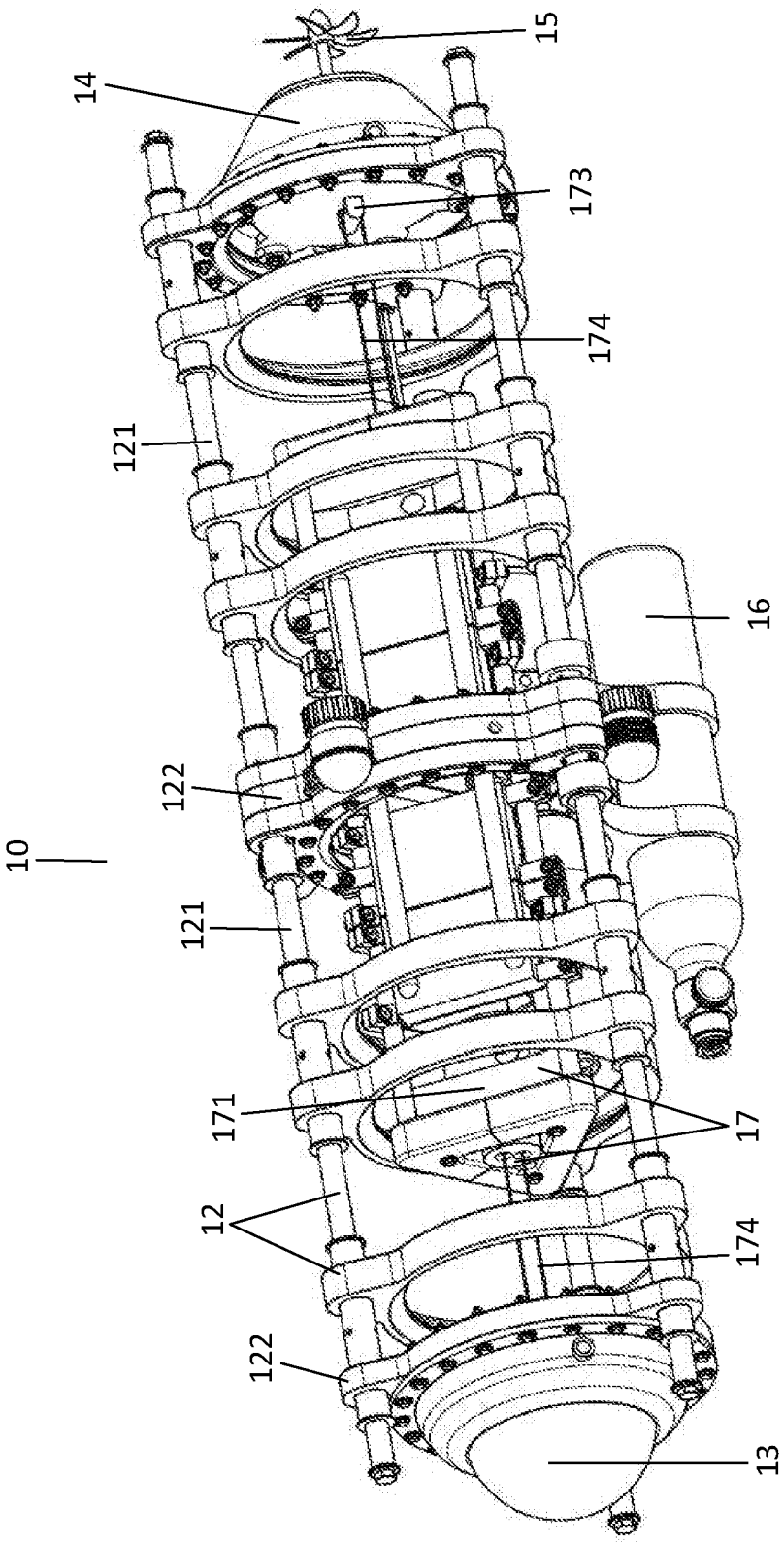


Fig. 3

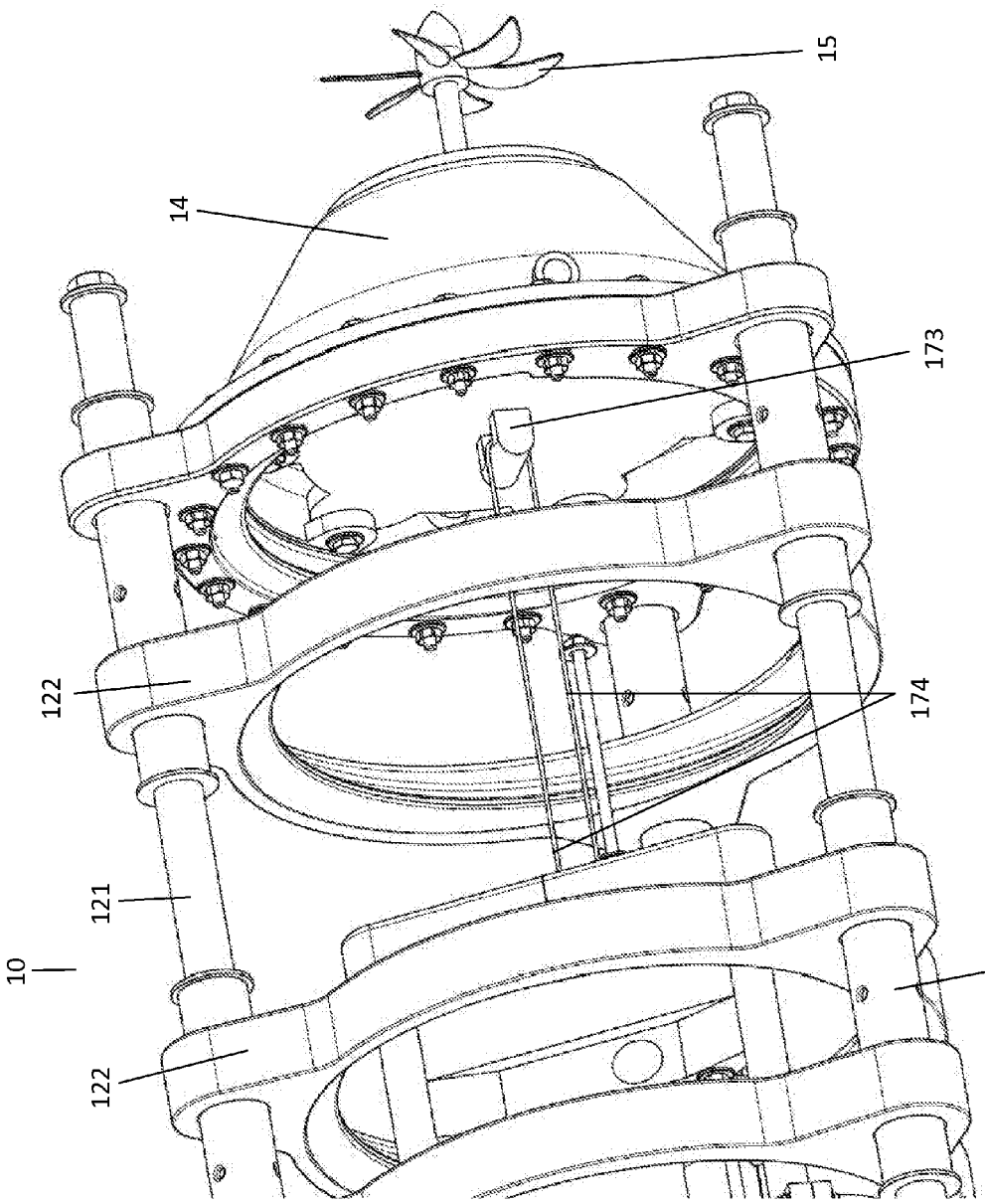


Fig. 4

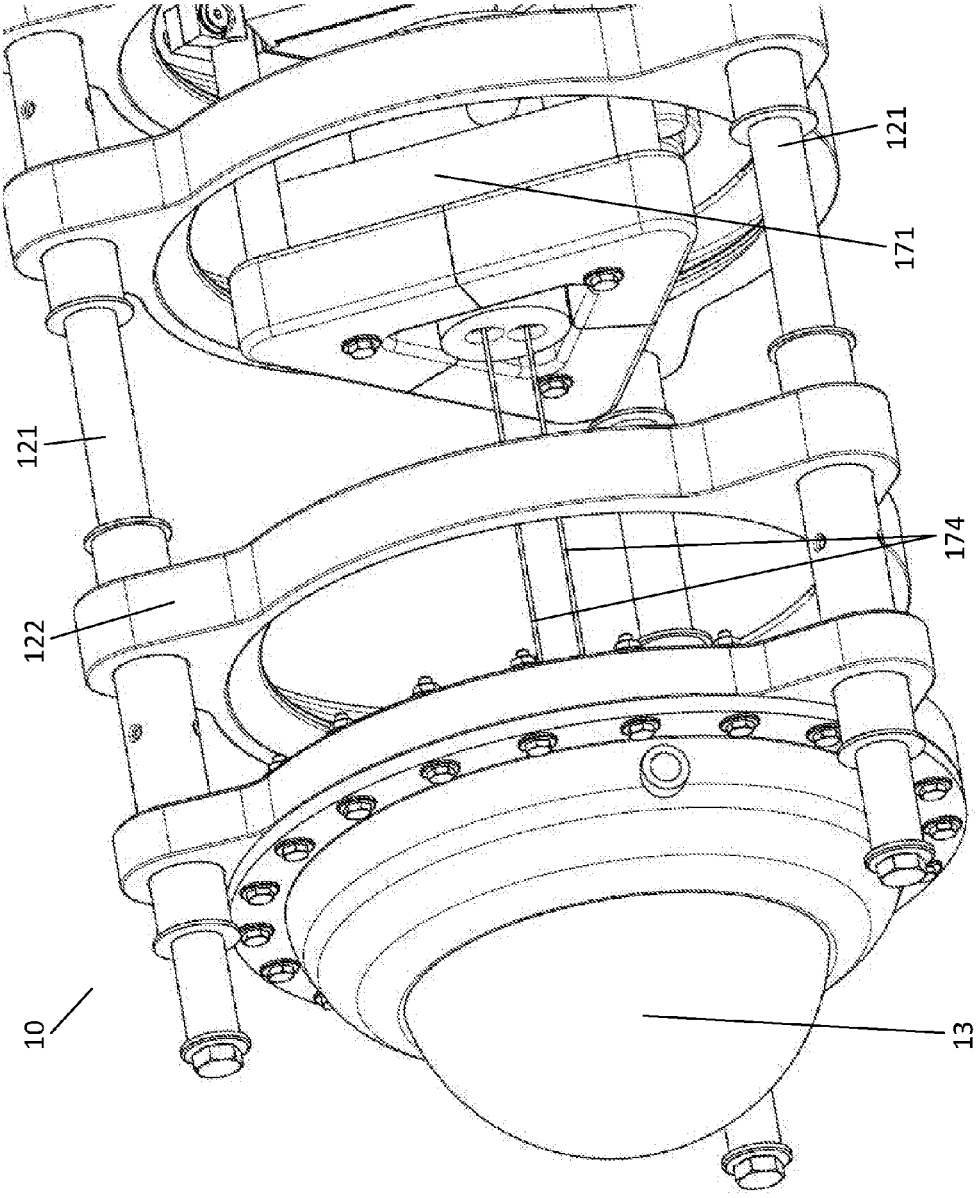


Fig. 5

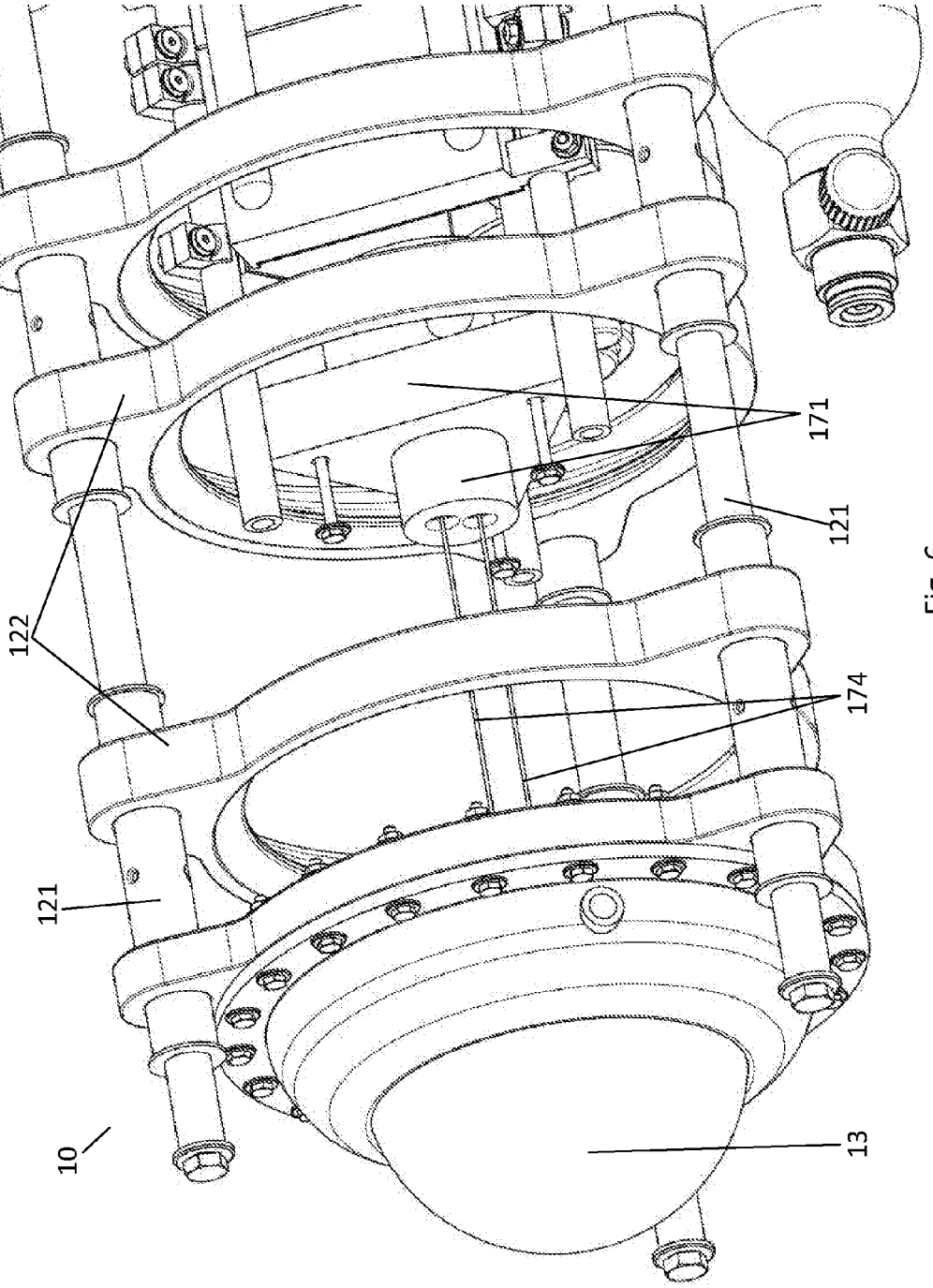


Fig. 6

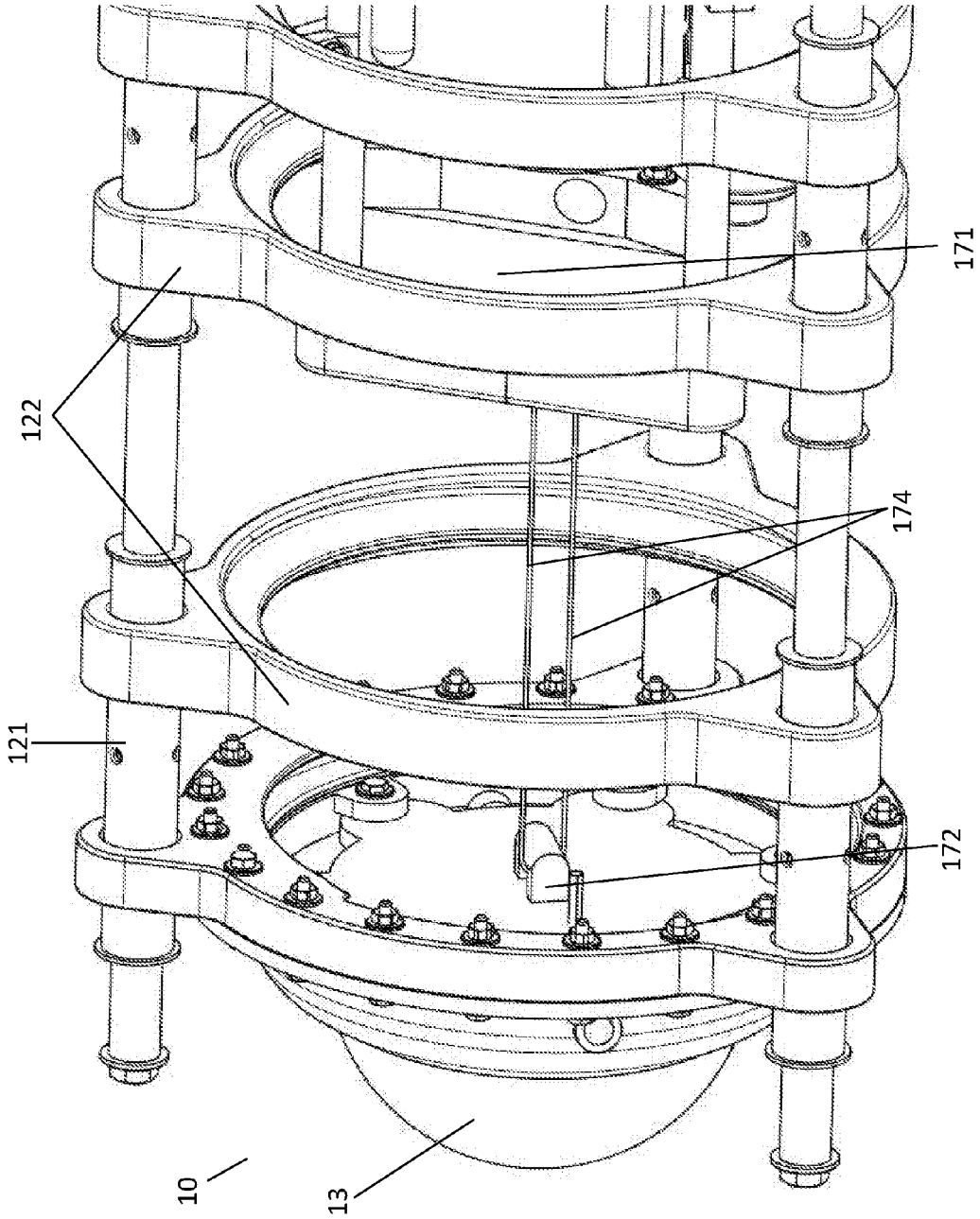


Fig. 7

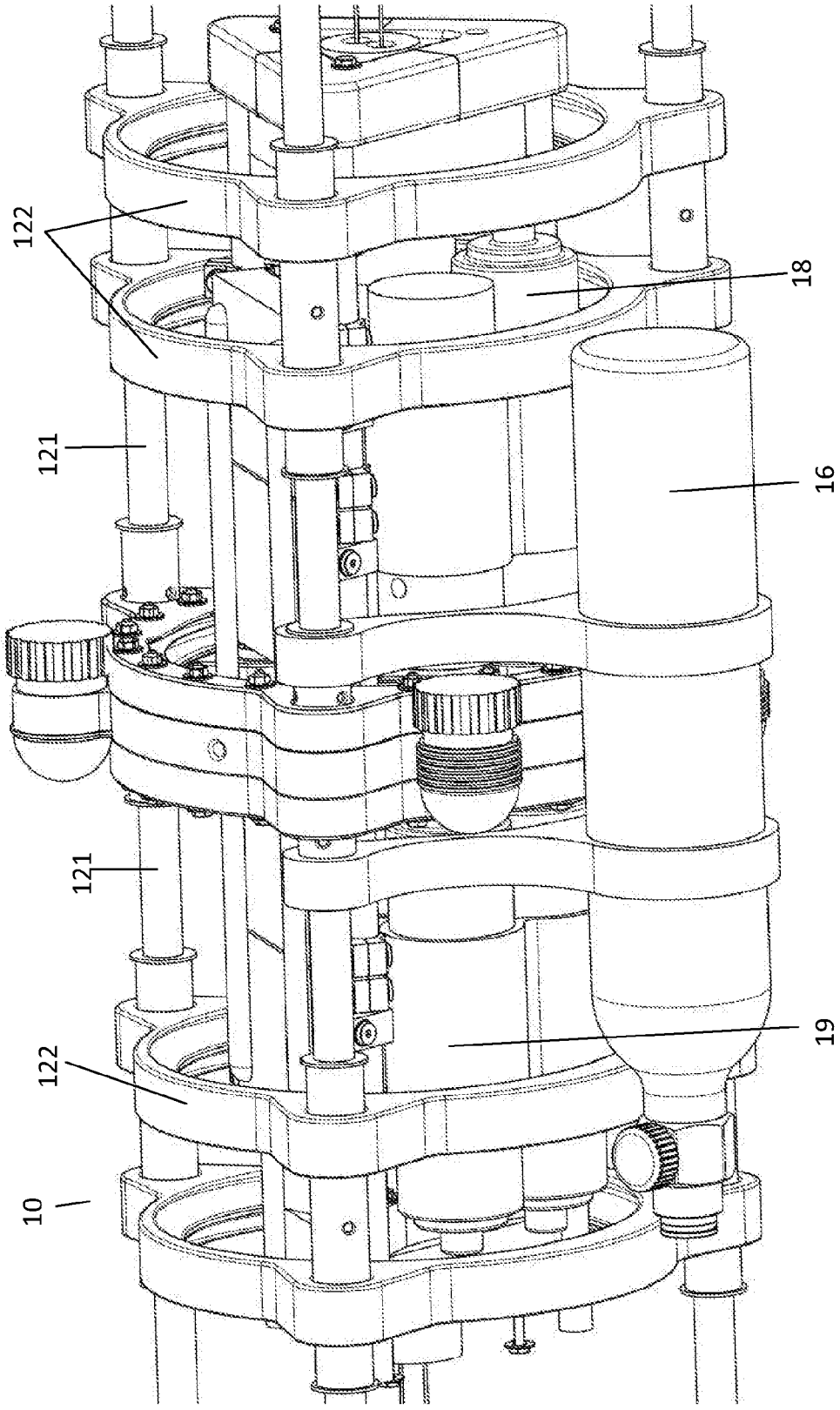


Fig. 8

UNMANNED UNDERWATER VEHICLE WITH VARIABLE-GEOMETRY HULL

TECHNICAL FIELD OF THE INVENTION

[0001] The invention provides an unmanned underwater vehicle with variable-geometry internally pressurized hull that enables the underwater vehicle to submerge/emerge and change submersion depth by varying hull's buoyancy and not the vehicle weight. Thus the vehicle needs not separate mechanisms for changing submersion depth, and the vehicle hull can be made lightweight; this provides for a structurally simple, lightweight, and cost-saving unmanned underwater vehicle, the payload of which can be substantially increased.

STATE OF THE ART AND BACKGROUND OF THE INVENTION

[0002] Known underwater vehicles destined to operate at varying submersion depths have rigid hulls of substantial strength, made of materials, the specific density of which is several times greater than the specific density of water. Such hulls are heavy, therefore the buoyancy/weight ratio is small, which substantially limits payload available. Moreover, steering machinery of known underwater vehicles needed to operate a vehicle at varying submersion depth includes ballast tanks, which are pressure vessels of considerable strength, hence heavy, pumps, plumbing and auxiliaries, which makes the system heavy, complicated, and costly.

[0003] Thus there is a need for an unmanned underwater vehicle of simple and lightweight construction and substantial payload.

SUMMARY OF THE INVENTION

[0004] The principal object of the instant invention is to provide a structurally simple unmanned underwater vehicle of lightweight structure.

[0005] A more specific object of the invention is to provide an unmanned underwater vehicle with simple submerging/emerging system without ballast tanks, pumps, plumbing and associated auxiliaries.

[0006] Yet more specific object of the invention is to provide an unmanned underwater vehicle with variable geometry hull, the buoyancy of which can be changed, thus enabling the vehicle to vary its submersion depth.

[0007] These and other objectives are achieved according to the invention by providing an unmanned underwater vehicle with internally pressurized, flexible hull, the buoyancy of which can be changed (preferably by changing its length), thus enabling the vehicle to vary its submersion depth. The flexible hull of the unmanned underwater vehicle according to the instant invention is filled with a gas, typically pressurized carbon dioxide, the pressure of which is adjusted so as to slightly exceed the pressure of water at given submersion depth, which allows the hull to be of exceptionally lightweight structure, and submersion depth steering system to be very simple, lightweight, and inexpensive.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] Preferred embodiment of the unmanned underwater vehicle is shown in accompanying drawings; like numerals refer to like vehicle's elements throughout all the figures.

[0009] FIG. **1** is a general view of a unmanned underwater vehicle according to the instant invention, where numeral **10** refers generally to the vehicle, numeral **11** refers to the vehicle flexible hull, numeral **12** refers to the vehicle external backbone; numeral **121** refers to longitudinal guides of transversal rings **122**; numeral **13** refers to a first cupola closing the hull at a first end, numeral **14** refers to a second cupola closing the hull at a second end, numeral **15** refers to propeller, and numeral **16** refers to pressurized carbon dioxide external tank.

[0010] FIG. **2** is another general view of the preferred embodiment of the unmanned underwater vehicle according to the present invention, exhibiting the flexible hull in a shortened configuration in comparison with the configuration shown in FIG. **1**.

[0011] FIG. **3** is a view of the preferred embodiment of the unmanned underwater vehicle according to the present invention with the flexible hull **11** removed showing internal arrangement of the vehicle, where numeral **17** refers generally to the flexible hull length control system, numeral **171** refers to an actuator, numeral **172** refers to a front pulley, numeral **173** refers to a rear pulley, and numeral **174** refers to a cable.

[0012] FIG. **4** is an enlarged view of the interior of the rear part of the preferred embodiment of the unmanned underwater vehicle according to the present invention.

[0013] FIG. **5** is an enlarged view of the interior of the front part of the preferred embodiment of the unmanned underwater vehicle according to the present invention.

[0014] FIG. **6** is another enlarged view of the interior of the rear part of the preferred embodiment of the unmanned underwater vehicle according to the present invention, with the fastener joining the actuator **171** to the flexible hull backbone **12**.

[0015] FIG. **7** is yet another enlarged view of the interior of the rear part of the preferred embodiment of the unmanned underwater vehicle according to the present invention, which shows the pulley **172** destined for adjusting the length of the front part of the flexible hull **11**.

[0016] FIG. **8** is an enlarged view of the interior of the central part of the preferred embodiment of the unmanned underwater vehicle according to the present invention, where numeral **18** refers to a evaporator of liquid carbon dioxide, and numeral **19** refers to an electric accumulator.

DETAILED DESCRIPTION OF A PREFERRED EMBODIMENT

[0017] A preferred embodiment of the unmanned underwater vehicle **10** according to the present invention, intended for illustrative purposes only, and not intended to limit the scope of the inventive ideas in any way, has a flexible hull **11** made of a flexible pipe closed at its front end by a first copula **13**, and closed at its rear end by a second copula **14**, where the flexible hull **11** is mounted on the external backbone **12** as described hereinafter. The external backbone **12** is composed of longitudinal guides **121**, and transverse rings **122** mounted slidingly on the longitudinal guides **121**. The flexible hull **11** of the preferred embodiment of the unmanned underwater vehicle **10** according to the invention is fixedly attached, e.g. by gluing, to transverse rings **122**, so as the flexible hull **11** length and thus the flexible hull **12** buoyancy can be varied by sliding the transverse rings **122** over the longitudinal guides **121**.

[0018] Exemplary flexible hull **10** length control system consists of at least one pneumatic, hydraulic, or electric actuator **171**, a first pulley **172** fixed at the front end of the

flexible hull **11**, a second pulley **173** fixed at the rear end of the flexible hull **11**, and a cable **174**. The cable **174** is wrapped around the pulleys **172** and **173**, and both the ends of the cable **174** are fixedly attached to the actuator **171**. The actuator **171** changes the effective length of the cable **174** wrapped around the pulleys **172**, **173** thus changing the distance between the cupolas **13** and **14** closing the flexible hull **11**, causing the length and buoyancy of the flexible hull **11** to vary, thus varying the submersion depth of the unmanned underwater vehicle **10**.

[0019] In order to balance the varying force exerted by water on the flexible hull **11** as the unmanned underwater vehicle **10** is submerged at varying depths, the vehicle **10** is equipped with a pressure balancing system that equates the pressure inside the flexible hull **11** with the pressure of ambient water. An exemplary pressure balancing system consists of an externally mounted liquefied carbon dioxide tank **16** fixedly attached to external backbone **12**, liquefied carbon dioxide vaporizer **18** placed inside the flexible hull **11**, suitable plumbing joining the liquefied carbon dioxide tank **16** with the liquefied carbon dioxide vaporizer **18** via a first steerable valve (not shown), the purpose of which is to convey a predetermined quantity of liquefied carbon dioxide from the tank **16** to vaporizer **18**, a second steerable valve (not shown) connecting the liquefied carbon dioxide vaporizer **18** with the interior of the flexible hull **11**, the purpose of which is to admit a predetermined quantity of pressurized carbon dioxide into the interior of the flexible hull so as to balance the pressure of water exerted on the flexible hull **11** as the vehicle submerge depth increases, and a third steerable valve (a wastegate, not shown) connecting the interior of the flexible hull **11** with ambient space, the purpose of which is to let off a predetermined quantity of pressurized carbon dioxide from the interior of the flexible hull to ambient space so as to balance the pressure of water exerted on the flexible hull **11** as the vehicle submerge depth decreases.

[0020] Placed inside the flexible hull **11** there is also a source of energy, preferably in the form of an accumulator (electric or hydraulic) **19**, the purpose of which is to supply power to the vehicle **10** various devices, including the actuator **171** driving the flexible hull **10** length varying system, main engine (not shown) driving the propeller **15**, and the pressure balancing system three main valves.

I claim:

1. Unmanned underwater vehicle having a variable buoyancy hull, wherein the variation of the hull buoyancy is attained by way of variation of the hull length.

2. Unmanned underwater vehicle according to claim **1**, wherein the variable buoyancy hull is made of a flexible pipe, wherein the length of said variable buoyancy hull made of flexible pipe can vary; wherein said variable buoyancy hull made of flexible pipe has a first end, a second end, and a longitudinal axis of symmetry of the variable buoyancy hull made of flexible pipe, wherein said first end of said variable buoyancy hull made of flexible pipe is closed by a first copula, and said second end of said variable buoyancy hull made of flexible pipe is closed by a second copula, so as said variable buoyancy hull made of a flexible pipe constitutes a watertight vessel.

3. Unmanned underwater vehicle according to claim **2**, wherein said unmanned underwater vehicle has an external backbone, wherein said external backbone consists of a first number K1 of longitudinal guides, and a second number K2 of transverse rings, wherein a third number K3<K1 of said second number K2 of said transverse rings are mounted slidingly on said first number K1 of said longitudinal guides; wherein each longitudinal guide of said first number K1 of said longitudinal guides extends parallel to said longitudinal axis of symmetry of the variable buoyancy hull made of flexible pipe; wherein the plane of each transverse ring of said second number K2 of said transverse rings is perpendicular to said longitudinal axis of symmetry of the variable buoyancy hull made of flexible pipe; wherein each transverse ring of said second number K2 of transverse rings is attached fixedly to said variable buoyancy hull made of a flexible pipe, wherein the first transverse ring of said second number K2 of transverse rings is attached fixedly to said first end of said variable buoyancy hull made of a flexible pipe, and the last transverse ring of said second number K2 of transverse rings is attached fixedly to said second end of said variable buoyancy hull made of a flexible pipe, wherein the transverse ring of said second number K2 of said transverse rings bearing a number k=2, . . . , K2−1 is placed between the transverse ring of said second number K2 of said transverse rings bearing a number k−1=1, 2, . . . , K2−1 and the transverse ring of said second number K2 of said transverse rings bearing a number k+1=3, . . . , K2 measured along said longitudinal axis of the variable buoyancy hull made of flexible pipe.

4. Unmanned underwater vehicle according to claim **3**, wherein a length-varying mechanism of said variable buoyancy hull made of flexible pipe is placed in said variable buoyancy hull made of flexible pipe, wherein said length-varying mechanism of said variable buoyancy hull made of flexible pipe has at least a first member, and a second member, wherein said first member of said length-varying mechanism of said variable buoyancy hull made of flexible pipe is connected with said second member of said length-varying mechanism of said variable buoyancy hull made of flexible pipe in such a way that the distance between said first member of said length-varying mechanism of said variable buoyancy hull made of flexible pipe and said second member of said length-varying mechanism of said variable buoyancy hull made of flexible pipe can vary.

5. Unmanned underwater vehicle according to claim **4**, wherein said transverse rings of said second number K2 of said transverse rings are mounted slidingly on said first number K1 of said longitudinal guides placed along said longitudinal axis of symmetry of the variable buoyancy hull made of flexible pipe, wherein said first member of said length-varying mechanism of said variable buoyancy hull made of flexible pipe is attached fixedly to said first transverse ring of said second number K2 of said transverse rings, and said second member of said length-varying mechanism of said variable buoyancy hull made of flexible pipe is attached fixedly to said last transverse ring of said second number K2 of said transverse rings.

6. Unmanned underwater vehicle according to claim **5**, wherein said length-varying mechanism of said variable buoyancy hull made of flexible pipe comprises a fourth number K4 of linear or rotary pneumatic actuators, wherein each pneumatic actuator of said fourth number K4 of linear or rotary pneumatic actuators has at least a body, and a sliding piston or a rotary piston, wherein said body of at least one pneumatic actuator of said fourth number K4 of linear or rotary pneumatic actuators is mounted fixedly to at least one transverse ring of said second number K2 of said

transverse rings, and the piston of said at least one pneumatic actuator of said fourth number K4 of linear or rotary pneumatic actuators is mounted fixedly to another transverse ring of said second number K2 of said transverse rings.

7. Unmanned underwater vehicle according to claim **5**, wherein said length-varying mechanism of said variable buoyancy hull made of flexible pipe comprises a fifth number K5 of linear or rotary hydraulic actuators, wherein each hydraulic actuator of said fifth number K5 of linear or rotary hydraulic actuators has at least a body, and a sliding piston or a rotary piston, wherein said body of at least one hydraulic actuator of said fifth number K5 of linear or rotary hydraulic actuators is mounted fixedly to at least one transverse ring of said second number K2 of said transverse rings, and the piston of said at least one hydraulic actuator of said fifth number K5 of linear or rotary hydraulic actuators is mounted fixedly to another transverse ring of said second number K2 of said transverse rings.

8. Unmanned underwater vehicle according to claim **5**, wherein said length-varying mechanism of said variable buoyancy hull made of flexible pipe comprises a sixth number K6 of linear or rotary electric actuators, wherein each electric actuator of said sixth number K6 of linear or rotary electric actuators has at least a body, and a slider or a rotor, wherein said body of at least one electric actuator of said sixth number K6 of linear or rotary electric actuators is mounted fixedly to at least one transverse ring of said second number K2 of said transverse rings, and the slider or rotor of said at least one electric actuator of said sixth number K6 of linear or rotary electric actuators is mounted fixedly to another transverse ring of said second number K2 of said transverse rings.

9. Unmanned underwater vehicle according to claim **3**, wherein a pressure balancing system is mounted on said external backbone, wherein said pressure balancing system comprises at least a compressed or fluidized gas tank, a first valve connecting said compressed or fluidized gas tank with the interior of said variable buoyancy hull made of flexible pipe, and a second valve connecting the interior of said variable buoyancy hull made of flexible pipe with ambient space; wherein opening said first valve with closed second valve causes the filling of said variable buoyancy hull made of flexible pipe with ambient space with compressed gas, wherein the pressure of said compressed gas filling said variable buoyancy hull made of flexible pipe is such that said pressure of said compressed gas filling said variable buoyancy hull made of flexible pipe balances the pressure the ambient water exerts on said variable buoyancy hull made of flexible pipe; wherein opening said second valve with closed said first valve causes letting off of the compressed gas from the interior of said variable buoyancy hull made of flexible pipe to ambient space.

* * * * *